

Testes

Testicular germ cell tumour (KITLG, SPRY4, BAK1, DMRT1, TERT), testicular function, testosterone and SHBG, hydrocele and other testicular diseases

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F's genes (genotyping chip + local imputation). It is not a diagnosis or a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Testicular germ cell tumour has a low polygenic predisposition: the consensus of the scores with good coverage is at percentile **13.7**, with the 231-variant score at **3.5** and the 52-variant score at percentile 23.9 (the same as in the Male Fertility report). The classic **KITLG** marker (rs995030) lacks the risk allele (A/A), and so do **TERT** (two loci) and **TEX14/SEPTIN4**. Male testosterone has typical-to-high genetics (consensus at **58.0**; most recent genomic score at percentile 89.3).

Unfavourable / mixed. The burden of 63 testicular tumour loci is at percentile 74.9 (typical), because there are risk alleles in large-effect loci: a second **KITLG** marker has 2 alleles (C/C), **DMRT1** 2 alleles (A/A), **ATF7IP** 2 alleles, and **SPRY4** and **BAK1** 1 allele each. **FSHB -211T** is present in 1 copy (G/T): FSH about 15% lower and less stimulation of Sertoli cells (see Male Fertility). Hydrocele and other testicular diseases have a 5-locus burden at percentile 84.7 (PAX8 and INHBB with 1 allele).

What it means. Testicular tumours are rare and Sérgio F's polygenic risk is low according to the scores, with some large-effect loci present that leave the loci burden in the middle of the distribution. Testicular function has typical genetics: typical-to-high testosterone and genetically high SHBG (consensus at percentile 84.9), with slightly lower FSH. Monthly self-examination remains the most useful measure, because the tumour has a very high cure rate when caught early.

2. How to read this report

- **Data:** genotypes from Sérgio F's chip (GRCh38, + strand) and, when the SNP is not on the chip, locally imputed genotypes (Beagle 5.5, public panel of 30x whole genomes) with quality $DR2 \geq 0.8$. Below this threshold the SNP is treated as not found in the data.
- **Key variants:** 14 SNPs selected from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 12 from the chip, 2 imputed and 0 not found.
- **Strand:** all genotypes are on the + strand of GRCh38. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 1 is palindromic (A/T or C/G), resolved by the reference allele and the position. Where the literature uses the gene strand, the letters may differ from those in this report.
- **Polygenic scores (PGS):** 10 PGS Catalog scores calculated locally with 2,574,198 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentile ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **GWAS loci burden:** for each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in the GWAS Catalog (1 row per SNP, the most significant) are added up as a simple count of effect alleles (each allele counts 1). Sérgio F's sum is compared with that of the reference population ($n = 633$) using exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium with each other, so the burden is a complement to, not a substitute for, the polygenic scores. 1791 loci, 1409 with genotype.
- **Judgement:** each result is classed as Favourable or Unfavourable (or typical / no judgement). A genotype is a probability, not a destiny; almost all effects are small and add to lifestyle.
- **Evidence:** Strong = replicated and with a clear clinical effect; Moderate = consistent GWAS/meta-analyses, modest effect; Weak = small or purely mechanistic studies.

3. Polygenic scores

Consensus by trait. For each group, the consensus is the **median of the percentiles of the scores with effect-size coverage $\geq 75\%$** , all calibrated in the same reference population ($n = 633$). The genomic column is the median of the scores with $\geq 100,000$ variants. Scores with coverage $< 75\%$ are shown in the tables but do not enter the consensus.

Group	Consensus (percentile)	Scores with coverage $\geq 75\%$	Genomic scores (percentile)	Genomic scores
Testicular tumour	13.7	2 of 5	—	0
Testosterone (men)	58.0	3 of 3	69.2	2
SHBG	84.9	2 of 2	87.2	1

For testicular tumour, a higher percentile means more risk. For testosterone, a higher percentile is favourable; SHBG is descriptive (more SHBG = more bound and less free testosterone). Three testicular tumour scores have 17–72% coverage and are left out of the consensus.

Testicular tumour

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Testicular tumour	PGS000086 (Graff 2020, 52 variants)	48 of 52 (coverage 90%)	23.9	TYPICAL	typical
Testicular tumour	PGS001164 (Tanigawa 2022, 280 variants)	184 of 280 (coverage 72%)	7.9	LESS LIKELY	favourable
Malignant testicular tumour	PGS001806 (Privé 2022, 231 variants)	223 of 231 (coverage 99%)	3.5	LESS LIKELY	favourable
Malignant testicular tumour	PGS000597 (Fritsche 2020, 771 variants)	142 of 771 (coverage 17%)	34.1	TYPICAL	typical
Malignant testicular tumour	PGS000600 (Fritsche 2020, 250 variants)	45 of 250 (coverage 20%)	7.4	LESS LIKELY	favourable

Testosterone (men)

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Testosterone (men, genomic)	PGS003559 (PGS Catalog 2023, 980,000 variants)	972,627 of 979,739 (coverage 99%)	89.3	HIGH	favourable
Testosterone (men, genomic)	PGS002205 (Privé 2022, 585,000 variants)	574,696 of 584,991 (coverage 97%)	49.0	TYPICAL	typical
Testosterone (men)	PGS001988 (Privé 2022, 3,985 variants)	3,857 of 3,985 (coverage 97%)	58.0	TYPICAL	typical

SHBG

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
SHBG (genomic)	PGS003552 (PGS Catalog 2023, 980,000 variants)	972,627 of 979,739 (coverage 99%)	87.2	HIGH	no judgement
SHBG	PGS001977 (Privé 2022, 51,000 variants)	49,749 of 51,427 (coverage 97%)	82.6	HIGH	no judgement

4. Key variants by topic

Replicated loci for testicular germ cell tumour (KITLG, SPRY4, BAK1, DMRT1, TERT, TEX14, ATF7IP), SHBG and FSH, and other testicular diseases (hydrocele). Links to the Male Fertility, Hormones and Andropause, and Reproductive Ageing reports.

Testicular germ cell tumour

Genes for germ cell survival and differentiation; some loci have effects of 1.5–2.5× per allele.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
KITLG	rs995030		A/A	chip	G (0 copies)	favourable	KITLG (KIT ligand, survival of germ cells): the G allele gives ~2.5× more testicular tumour per copy; the largest locus	Strong
KITLG	rs3782181		C/C	chip	C (2 copies)	unfavourable	KITLG: second marker of the same locus [GWAS Catalog: Testicular germ cell tumor, allele C, OR 2.07, p=3E-129, PMID 28604728]	Strong
SPRY4	rs4624820		G/A	chip	G (1 copy)	unfavourable (mild)	SPRY4 (brake on the KIT/MAPK pathway): ~1.5× per copy [GWAS Catalog: Testicular germ cell tumor, allele G, OR 1.47, p=3E-57, PMID 28604728]	Strong
BAK1	rs210138		A/G	chip	G (1 copy)	unfavourable (mild)	BAK1 (apoptosis of germ cells): ~1.5× per copy [GWAS Catalog: Testicular germ cell tumor, allele G, OR 1.48, p=4E-37, PMID 28604728]	Strong
DMRT1	rs7040024		A/A	chip	A (2 copies)	unfavourable	DMRT1 (determination and maintenance of the testis): ~1.5× per copy [GWAS Catalog: Testicular germ cell tumor, allele A, OR 1.53, p=2E-45, PMID 28604728]	Strong
TERT	rs4635969		G/G	chip	A (0 copies)	favourable	TERT/CLPTM1L (5p15): ~1.5× per copy [GWAS Catalog: Testicular germ cell tumor, allele A, OR 1.46, p=3E-27, PMID 28604732]	Moderate
TERT	rs2736100		C/C	chip	A (0 copies)	favourable	TERT intron 2: telomere length and testicular tumour [GWAS Catalog: Testicular germ cell tumor, allele A, OR 1.28, p=9E-25, PMID 28604728]	Moderate
SEPTIN4/TEX14	rs9905704		T/T	chip	G (0 copies)	favourable	TEX14/SEPTIN4 (intercellular bridges of germ cells) [GWAS Catalog: Testicular germ cell tumor, allele G, OR 1.27, p=3E-20, PMID 28604728]	Moderate
ATF7IP	rs2900333		C/C	chip	C (2 copies)	unfavourable	ATF7IP (12p13, epigenetic regulation) [GWAS Catalog: Testicular germ cell tumor, allele C, OR 1.32, p=1E-13, PMID 23666240]	Moderate

Testicular function, testosterone and SHBG

SHBG and FSH: testosterone transport and Sertoli cell stimulation (no judgement).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SHBG	rs12150660		G/T	chip	T (1 copy)	no judgement	SHBG (globulin transporting testosterone): levels of SHBG [GWAS Catalog: Sex hormone-binding globulin levels, allele T, beta 6.140 [5.83-6.45] nmol/L increase, p=1E-300, PMID 34321204]	Strong

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SHBG	rs6258	P156L	C/C	chip	C (2 copies)	no judgement	P156L of SHBG: less affinity for testosterone [GWAS Catalog: Total testosterone levels, allele C, beta 0.718 [0.68-0.75] unit increase, p=6E-347, PMID 32042192]	Moderate
FSHB	rs10835638		G/T	chip	T (1 copy)	no judgement	–211G>T of FSHB: the T allele lowers FSH (less stimulation of Sertoli cells and sperm production)	Moderate

Development and other testicular diseases

Hydrocele and benign diseases of the testis and genital tract.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PAX8	rs4849183		G/C	imputed DR2=0.99	C (1 copy)	unfavourable (mild)	PAX8 (development of the genital tract): hydrocele and other testicular diseases [GWAS Catalog: Other disorders of testis (PheCode 603), allele G, beta 0.173 [0.14-0.2] unit decrease; risk/increase allele = C, the opposite, p=8E-27, PMID 39024449]	Moderate
INHBB	rs6718024		C/T	imputed DR2=1.00	T (1 copy)	unfavourable (mild)	INHBB (inhibin B, produced by Sertoli cells): hydrocele [GWAS Catalog: Other disorders of testis (PheCode 603), allele C, beta 0.145 [0.12-0.17] unit decrease; risk/increase allele = T, the opposite, p=9E-22, PMID 39024449]	Moderate

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 5 \times 10^{-8}$) for testicular tumour, testicular hypofunction, hydrocele and male testosterone. Alleles are oriented towards more risk (or towards higher testosterone) and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F's effect alleles	Calibrated variants	Percentile	Judgement
Testicular germ cell tumour	67 (63)	66 of 126	59	74.9	typical
Testicular hypofunction	6 (6)	7 of 12	5	59.9	typical
Hydrocele and other testicular diseases	6 (5)	7 of 10	5	84.7	unfavourable
Testosterone (men)	1712 (1335)	1289 of 2670	1331	38.4	typical

6. Favourable

- **Testicular tumour, low scores:** consensus at percentile **13.7**; 231-variant score at percentile 3.5 and 52-variant score at percentile 23.9. **Moderate.**
- **KITLG rs995030** without the risk allele (A/A); **TERT** (G/G, C/C) and **TEX14/SEPTIN4** (T/T) without risk alleles. **Moderate to strong.**
- **Male testosterone:** consensus at **58.0**; burden of 1,335 loci at percentile 38.4. **Weak to moderate.**
- **Testicular hypofunction:** loci burden at percentile 59.9 (typical). **Weak.**

7. Unfavourable

- **Large-effect testicular tumour loci:** KITLG rs3782181 C/C, DMRT1 A/A and ATF7IP C/C (2 alleles each), SPRY4 and BAK1 (1 allele); loci burden at percentile 74.9. **Moderate.**
- **FSHB –211G>T** G/T: FSH about 15% lower (Male Fertility report: FSH score at the 12th percentile). **Moderate.**
- **Genetically high SHBG** (consensus at percentile 84.9; SHBG rs12150660 G/T): less free testosterone per unit of total testosterone. **Weak.**
- **Hydrocele and other testicular diseases:** 5-locus burden at percentile 84.7. **Weak.**

8. Mechanisms of ageing

Mapping onto the 84 ageing mechanisms of Sérgio F's immortality goals (only the mechanisms touched by this report).

Mechanism	Genetic finding	Meaning
Endocrine Dysregulation	Testicular axis: typical-to-high testosterone, high SHBG, slightly lower FSH	Typical
Somatic Mutation Accumulation	Germ cell tumour: low scores (consensus 13.7), some large-effect loci present	Favourable
Telomere Attrition and Hayflick Limit	TERT without the testicular tumour risk alleles (the same loci regulate telomere length)	Typical
Stem Cell Exhaustion	Spermatogonia are the stem cells of the testis; KITLG/KIT and DMRT1 regulate their survival	Conditional
Lean Mass Loss	Testosterone is the main anabolic signal for muscle in men; typical-to-high genetics	Favourable

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Monthly testicular self-examination** (after a shower): look for hard lumps, enlargement or a feeling of heaviness; any change justifies a scrotal ultrasound. *Suggestion (not applied)*.
2. **Total testosterone, SHBG and calculated free testosterone**, in the morning, if there are symptoms or with routine blood tests: genetically high SHBG can lower the free fraction. *Suggestion (not applied)*.
3. **Testicular heat and fertility**: regular sauna use temporarily lowers sperm count; if fatherhood is planned, pause it 3 months beforehand. *Suggestion (not applied)*.
4. **Rapamycin and spermatogenesis** (Male Fertility report): if there are plans to have children, discuss sperm cryopreservation or a 3-month pause with the doctor. *Suggestion (not applied)*.
5. **Avoid endocrine disruptors** (heated plastics, pesticides): the testis is one of the most sensitive organs. *Suggestion (not applied)*.

10. Limitations

- The testicular tumour scores have few variants; three have low coverage and are left out of the consensus.
- The large-effect loci (KITLG, DMRT1) have markers with different readings (rs995030 without risk, rs3782181 with 2 alleles), which shows that the linkage disequilibrium between them is incomplete.
- The chip does not read Y chromosome microdeletions, androgen receptor variants (CAG repeat) or rare mutations affecting testicular function.
- Testosterone depends heavily on sleep, training and medicines; genetics explains a modest part.
- The loci burden adds up SNPs in linkage disequilibrium; it is descriptive.
- Conclusions from genes only; this is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F's chip (GRCh38, + strand). Missing SNPs come from local imputation (Beagle 5.5, public panel of 30x whole genomes), used only with $DR2 \geq 0.8$. No genetic data left the analysis computer.
- **Alleles and strand:** the published effect allele is compared with the alleles of the imputed VCF; for palindromic SNPs the + strand is assumed (the GWAS Catalog and PGS Catalog convention). Published alleles with a lowering effect were inverted to the raising direction.
- **Polygenic scores:** GRCh38 harmonised files from the PGS Catalog (10 scores: testicular tumour, testosterone and SHBG). Score = $\sum \text{coefficient} \times \text{dosage}$. The percentile is empirical in the reference population ($n = 633$), using the same variants. Consensus = median of the scores with effect-size coverage $\geq 75\%$.
- **Loci burden:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), 4 sets, alleles oriented by the sign of the published effect, with empirical calibration in the same reference population.
- **Interpretation:** Rapley 2009 and Kanetsky 2009 (KITLG and SPRY4), Pluta 2021 (germ cell tumour GWAS), Ruth 2020 (testosterone and SHBG), Grigorova 2008 (FSHB –211G>T), PGS Catalog, GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Strand	Interpretation
Testicular germ cell tumour	KITLG	rs995030	A/A	chip	G	0	+ strand	without the risk allele
Testicular germ cell tumour	KITLG	rs3782181	C/C	chip	C	2	+ strand	2 copies of the risk allele
Testicular germ cell tumour	SPRY4	rs4624820	G/A	chip	G	1	+ strand	1 copy of the risk allele
Testicular germ cell tumour	BAK1	rs210138	A/G	chip	G	1	+ strand	1 copy of the risk allele
Testicular germ cell tumour	DMRT1	rs7040024	A/A	chip	A	2	+ strand	2 copies of the risk allele
Testicular germ cell tumour	TERT	rs4635969	G/G	chip	A	0	+ strand	without the risk allele
Testicular germ cell tumour	TERT	rs2736100	C/C	chip	A	0	+ strand	without the risk allele
Testicular germ cell tumour	SEPTIN4/TEX14	rs9905704	T/T	chip	G	0	+ strand	without the risk allele
Testicular germ cell tumour	ATF7IP	rs2900333	C/C	chip	C	2	+ strand	2 copies of the risk allele
Testicular function, testosterone and SHBG	SHBG	rs12150660	G/T	chip	T	1	+ strand	1 copy of the effect allele
Testicular function, testosterone and SHBG	SHBG	rs6258	C/C	chip	C	2	+ strand	2 copies of the effect allele
Testicular function, testosterone and SHBG	FSHB	rs10835638	G/T	chip	T	1	+ strand	1 copy of the effect allele
Development and other testicular diseases	PAX8	rs4849183	G/C	imputed DR2=0.99	C	1	+ strand (palindromic; + assumed)	1 copy of the risk allele
Development and other testicular diseases	INHBB	rs6718024	C/T	imputed DR2=1.00	T	1	+ strand	1 copy of the risk allele